Proof

Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k},\ \vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k},\ \vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$. Then, we have

$$\vec{a},\vec{b},\vec{c} \text{ are coplanar } \Leftrightarrow \left[\vec{a},\vec{b},\vec{c}\right] = 0 \Leftrightarrow \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = 0$$

$\Leftrightarrow$ there exist scalars $r,s,t \in \mathbb{R}$,

atleast one of them non-zero such that

$$a_1r + a_2s + a_3t = 0,\ b_1r + b_2s + b_3t = 0,\ c_1r + c_2s + c_3t = 0$$

$\Leftrightarrow$ there exist scalars $r,s,t \in \mathbb{R}$,

atleast one of them non-zero such that $r\vec{a} + s\vec{b} + t\vec{c} = \vec{0}$.

Theorem 6.6

If $\vec{a},\vec{b},\vec{c}$ and $\vec{p},\vec{q},\vec{r}$ are any two systems of three vectors, and if $\vec{p} = x_1\vec{a} + y_1\vec{b} + z_1\vec{c}$, $\vec{q} = x_2\vec{a} + y_2\vec{b} + z_2\vec{c}$, and, $\vec{r} = x_3\vec{a} + y_3\vec{b} + z_3\vec{c}$, then

$$\left[\vec{p},\vec{q},\vec{r}\right] = \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} \left[\vec{a},\vec{b},\vec{c}\right].$$

Note

By theorem 6.6, if $\vec{a},\vec{b},\vec{c}$ are non-coplanar and

$$\begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} \neq 0,$$

then the three vectors $\vec{p} = x_1\vec{a} + y_1\vec{b} + z_1\vec{c}$, $\vec{q} = x_2\vec{a} + y_2\vec{b} + z_2\vec{c}$, and, $\vec{r} = x_3\vec{a} + y_3\vec{b} + z_3\vec{c}$ are also non-coplanar.

Example 6.12

If $\vec{a} = -3\hat{i} - \hat{j} + 5\hat{k},\ \vec{b} = \hat{i} - 2\hat{j} + \hat{k},\ \vec{c} = 4\hat{j} - 5\hat{k}$, find $\vec{a}\cdot(\vec{b}\times\vec{c})$.

Solution: By the defination of scalar triple product of three vectors,

$$\vec{a}\cdot(\vec{b}\times\vec{c}) = \begin{vmatrix} -3 & -1 & 5 \\ 1 & -2 & 1 \\ 0 & 4 & -5 \end{vmatrix} = -3.$$

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